

Genetic Programming Guided Local Search for Large-scale Vehicle Routing Problem

Presenter: Saining Liu

Supervisors: A/Prof. Yi Mei, Prof. Mengjie Zhang



Content

1 Background

2 GPGLS

Framework

Experiments

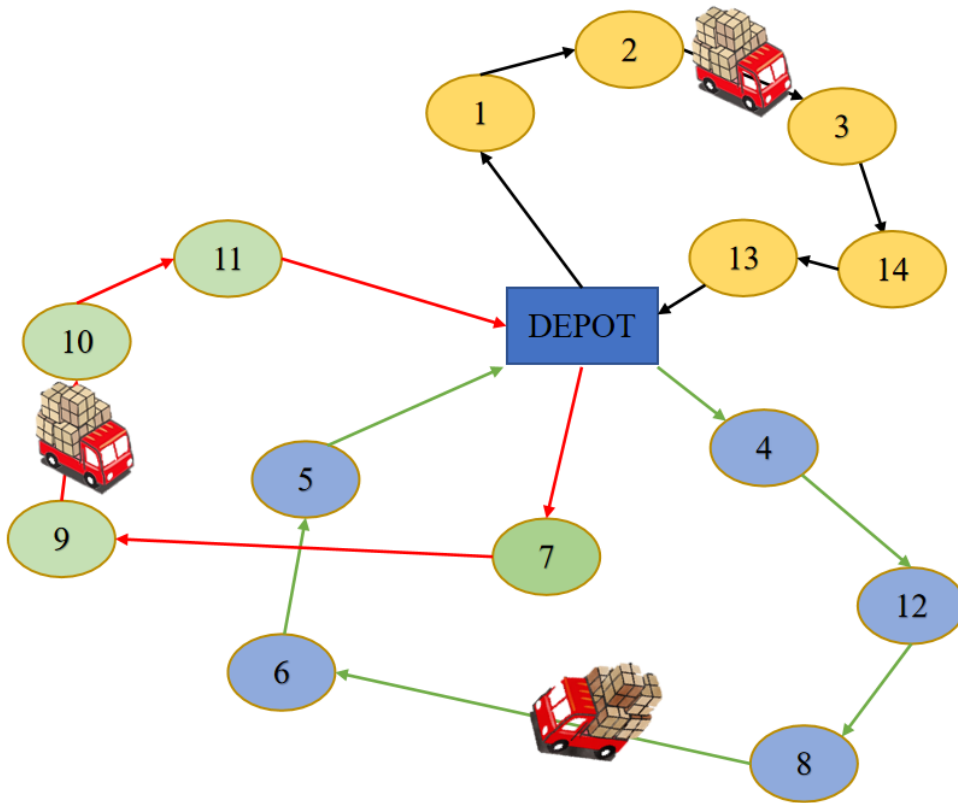
Analysis

Conclusion

1 BACKGROUND

Background

The **Vehicle Routing Problem (VRP)** is crucial in the field of **logistics and transportation**, which aims to **find routes** which serve a set of customers, **optimizing the costs** involved.



- Enhancing **environmental** sustainability
- Elevating **service** efficacy
- mitigating **traffic** congestion

The VRP was formally defined as a graph $G = (V, A)$ where $V = \{0, 1, \dots, n\}$ denotes the set of **nodes** (or vertices) within the graph, $A = \{(i, j) | i, j \in V, i \neq j\}$ denotes the set of **edges** that connect these nodes.

Background

Capacitated VRP revolves around optimizing the routes of a limited fleet of vehicles to efficiently serve a set of customers while adhering to the **capacity constraint**.

The inputs:

- A set of $(n + 1)$ points, consisting of **one depot** and n customers.
- A $(n + 1) \times (n + 1)$ matrix $d = [d_{ij}]$ containing **distances** between each pair of points i and j .
- An n -dimensional **demand vector** specifying the quantity to be delivered to each customer i .
- A vehicle **capacity** Q .

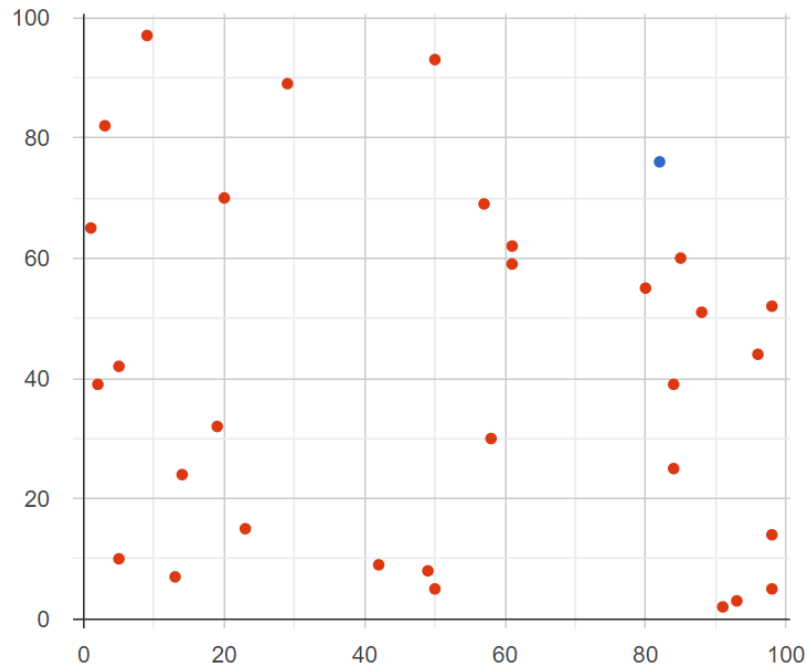
The outputs:

- A solution includes **set of routes**, which should **start and end at the depot**.
- Every customer is served **exactly once**.
- The **sum demands** of one route **cannot exceed** Q .

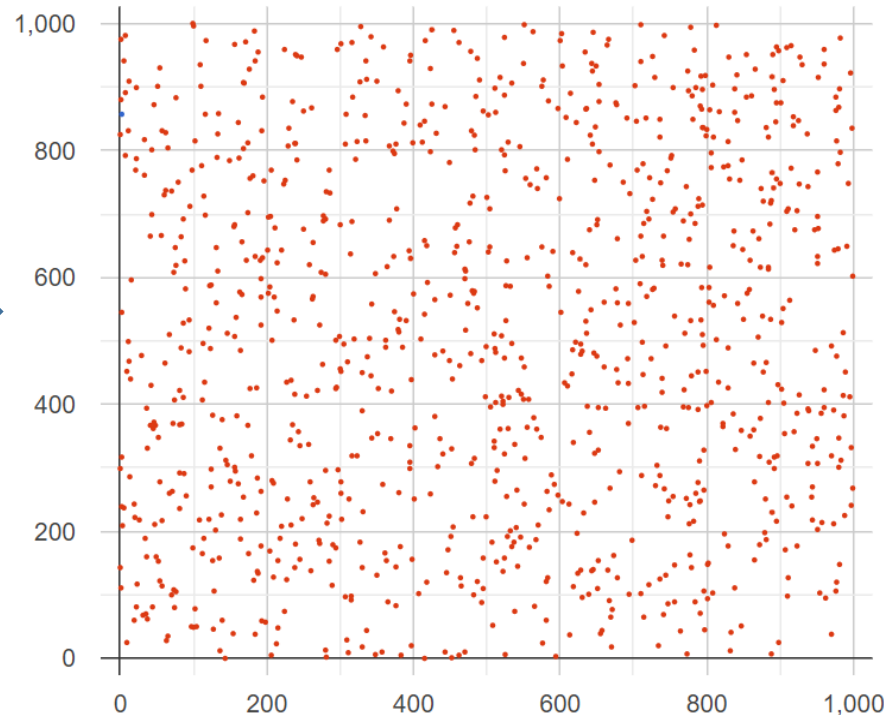
Background

Large-scale VRP (LSVRP) is a special case of the CVRP when the number of customers is more than **200**.

A-n32-k5 (n=31, Q=100)



X-n1001-k43 (n=1000, Q=131)



NP-hard problem [1]

The solution space of the CVRP and LSVRP **grows exponentially** with the number of nodes.

Due to the larger scale and higher complexity of the LSVRP, the methods should be **more efficient and flexible**.

[1] Jan Karel Lenstra and AHG Rinnooy Kan. Complexity of vehicle routing and scheduling problems. *Networks*, 11(2):221–227, 1981.

2 GPGLS

```
procedure Guided Local Search ( $p, g, \lambda, [I_1, \dots, I_M], [c_1, \dots, c_M], M$ )
begin
   $k \leftarrow 0$ ;
   $s_0 \leftarrow \text{ConstructionMethod}(p)$ ;
  /*set all penalties to 0 */
  for  $i \leftarrow 1$  until  $M$  do
     $p_i \leftarrow 0$ ;
    /* define the augmented objective function */
     $h \leftarrow g + \lambda * \sum p_i * I_i$ ;
  while Stopping Criterion do
    begin
       $s_{k+1} \leftarrow \text{ImprovementMethod}(s_k, h)$ ;
      /* compute the utility of features*/
      for  $i \leftarrow 1$  until  $M$  do
         $util_i \leftarrow I_i(s_{k+1}) * \frac{c_i}{1+p_i}$ ;
        /* penalize features with maximum utility */
        for each  $i$  such that  $util_i$  is maximum do
           $p_i \leftarrow p_i + 1$ ;
         $k \leftarrow k + 1$ ;
      end
     $s^* \leftarrow$ best solution found with respect to objective function  $g$ ;
    return  $s^*$ ;
  end
```

GLS augments the given objective function with penalties.

The conceptual idea of GLS is to change the cost evaluation of s^* , rather than s^* itself.

Features that are considered bad are penalized.

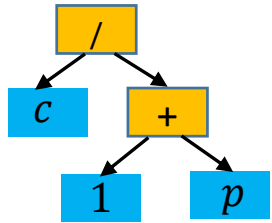
In the context of the VRP, those features are edges, and the costs of bad edges are increased [5].

p	The problem
g	The objective function
h	The augmented objective function
λ	A parameter
I_i	The indicator function of feature i
c_i	The cost of feature i
M	The number of features
p_i	The penalty of feature i

[5] Florian Arnold and Kenneth Sørensen. Knowledge-guided local search for the vehicle routing problem. Computers & Operations Research, 105:32–46, 2019.

GPGLS-Framewok

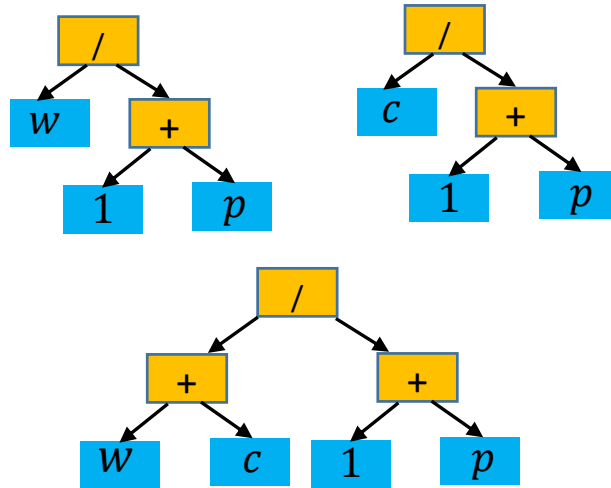
$$util = \frac{c}{1+p}$$



GLS

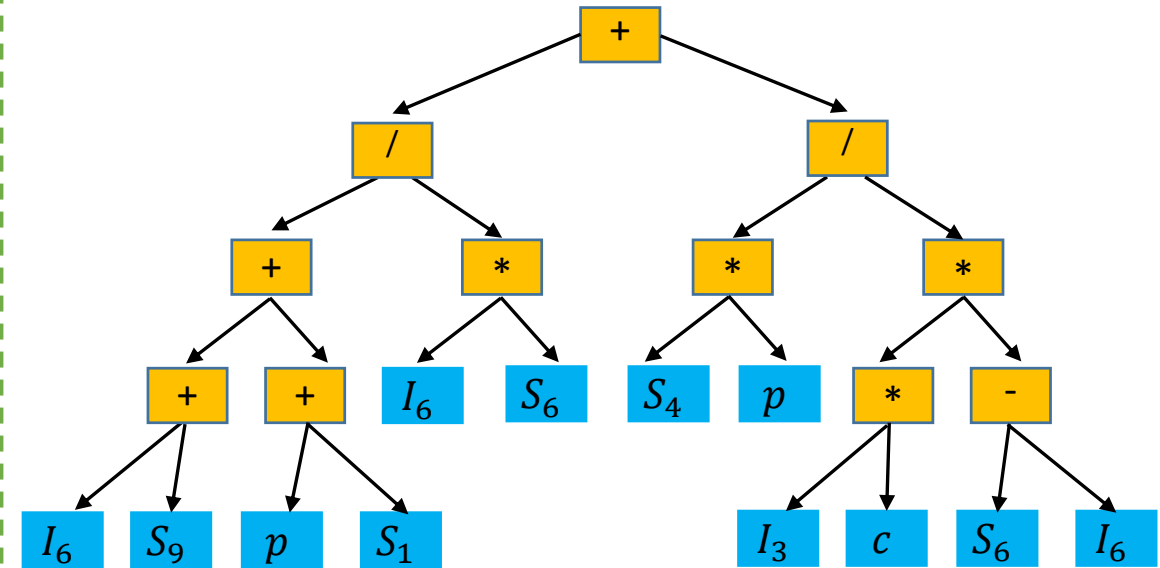
$$util = \frac{c}{1+p} \quad util = \frac{w}{1+p}$$

$$util = \frac{w+c}{1+p}$$



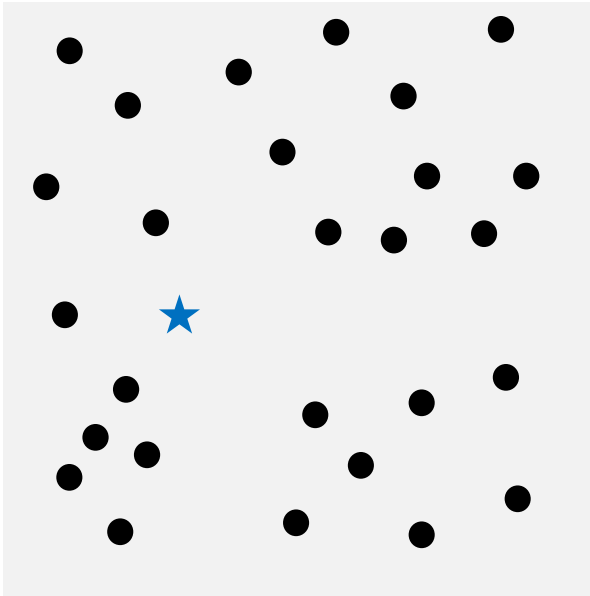
Knowledge GLS (KGLS)

$$util = \left(\frac{(I_6 + S_9) + (p + S_1)}{I_6 * S_6} \right) + \left(\frac{S_4 * p}{(I_3 * p) * (S_6 - I_6)} \right)$$



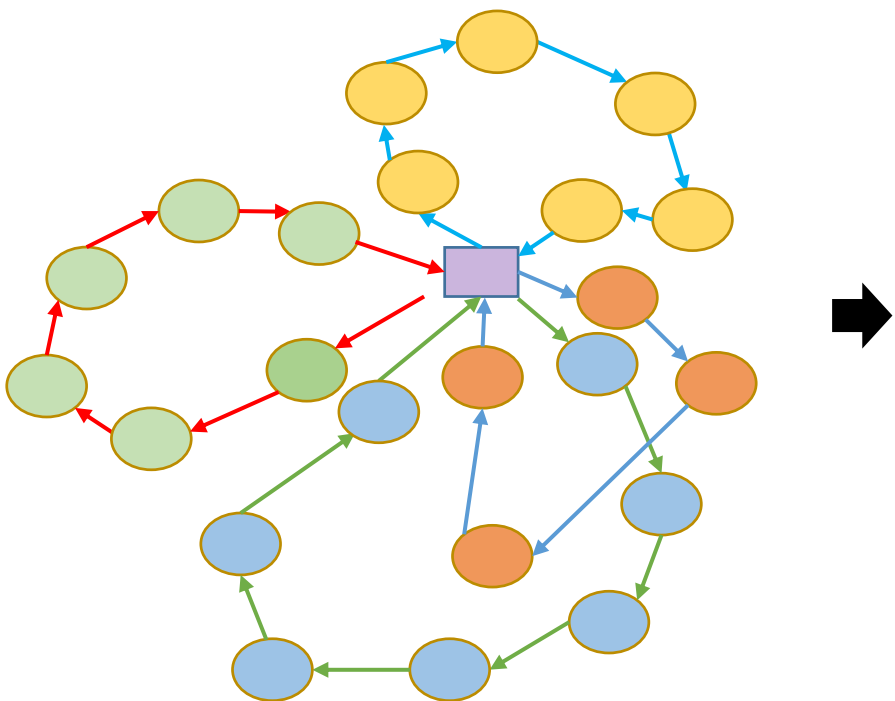
GPGLS

Terminals based on factors of instances



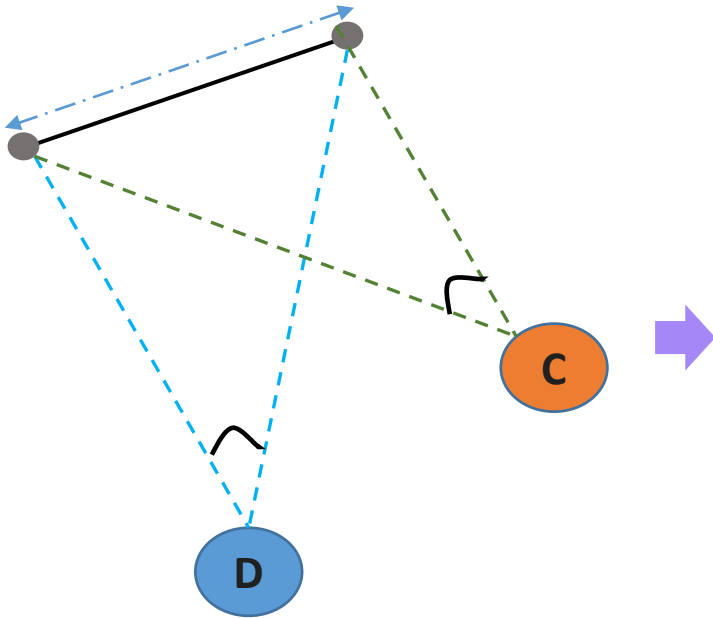
I1	Number of customers.
I2	Number of routes.
I3	Degree of capacity utilization.
I4	Average distance between each pair of customers.
I5	Standard deviation of the pairwise distance between customers.
I6	Average distance from customers to the depot.
I7	Standard deviation of the distance from customers to the depot.
I8	Standard deviation of the radians of customers towards the depot.

Terminals based on factors of solutions



S1	Average number of intersections per customer.
S2	Longest distance between two connected customers per route.
S3	Average distance between depot to directly-connected customers.
S4	Average distance between their centers of gravity.
S5	Average width per route.
S6	Average span in radian per route.
S7	Average compactness per route, measured by width.
S8	Average compactness per route, measured by radian.
S9	Average depth per route.
S10	Standard deviation of the number of customers per route.
S11	Number of feasible routes in current solution.
S12	Average cost of the routes in current solution.

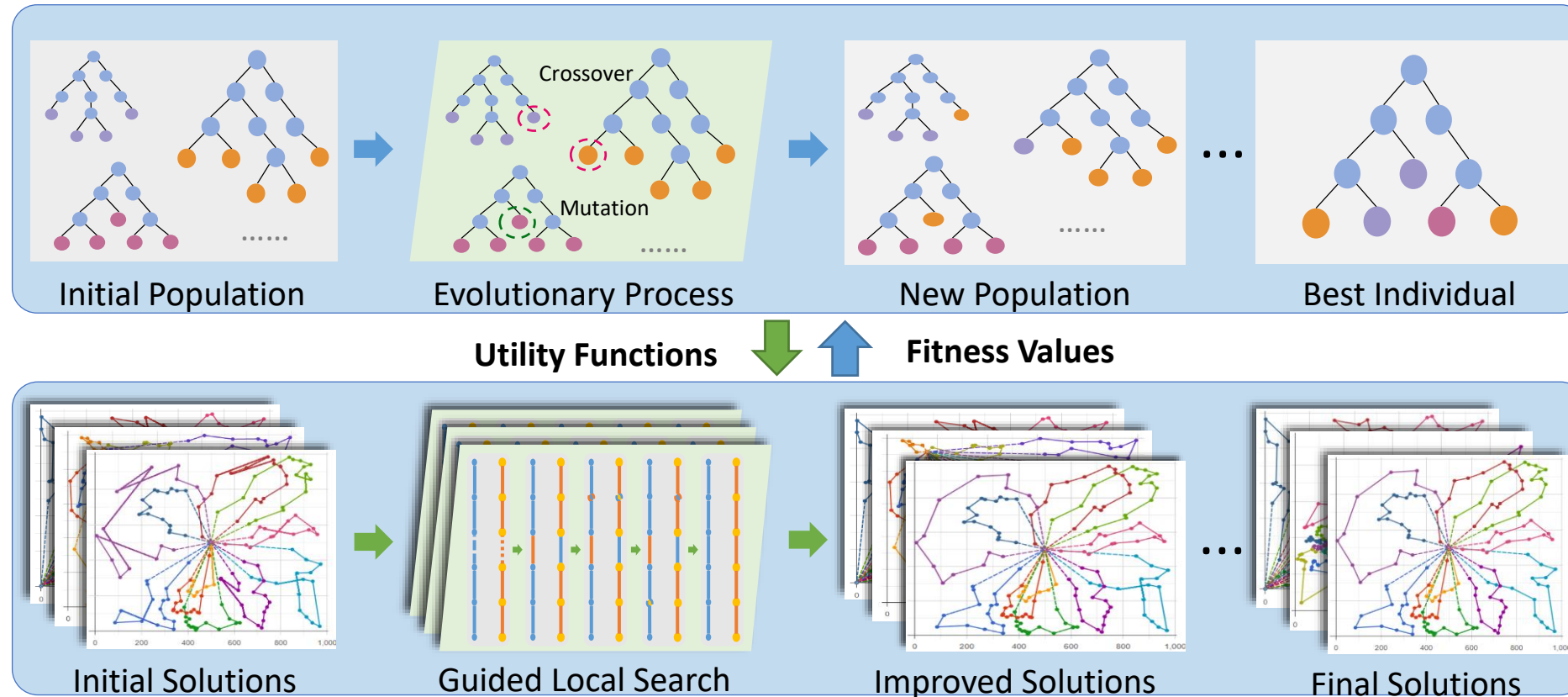
Terminals based on factors of edges.



c	Euclidean distance of the two end-nodes of the edge.
p	The number of times the edge has been penalized.
w	How far the center of gravity is positioned from the edge with respect to their distances from depot.
θ_h	The angle between the two end-nodes of the edge at the horizontal axis, ranging within $[0, 2\pi]$.
θ_c	The variation in angles between the two end-nodes of the edge with respect to the central point, ranging within $[0, 2\pi]$.

GPGLS-Framework

The Framework of GPGLS



GPGLS: GP + GLS.

The upper part is the **GP evolutionary process**, while the lower part is the **GLS-based fitness evaluator**.

GPGLS-Experiments

Experiements

Parameters	Settings
Population size	100
Generations	50
Maximal tree depth	6
Crossover probability	0.8
Mutation probability	0.2
Elitism	2

Train	Test
21 instances	79 instances
< 200 customers	$\geq$ 200 customers
5 seconds	10 minutes

In order to get a high-quality solution, we have to allocate **enough time to do local search**. However, **the efficiency of training** process is also important. In order to **keep a balance** of them and make the fitness between training and test **similar**, we use **small-scale instances and little time** in the training process.

GPGLS-Experiments

Experiment results

BKS: the cost of best-known solution

CW: the cost CW heuristic solution

GLS: the cost of GLS solution

KGLS: the cost of KGLS solution

Gap: $\frac{GPGLS - BKS}{BKS} * 100$

Sig-Test: significant test between

GPGLS and KGLS.

Instances	BKS	CW	GLS	KGLS	GPGLS	Gap(%)	Sig-Test
X-n200-k36	58578	61167	58868	58747	58778	0.34	≈
X-n204-k19	19565	21271	19611	19666	19618	0.27	+
X-n209-k16	30656	32635	30760	30733	30698	0.14	+
X-n214-k11	10856	11816	11127	10919	10924	0.63	≈
X-n219-k73	117595	118364	117775	117674	117753	0.13	+
X-n223-k34	40437	42357	40886	40675	40616	0.44	+
X-n228-k23	25742	27198	26497	25838	25882	0.54	≈
X-n233-k16	19230	20433	19413	19333	19342	0.58	-
X-n237-k14	27042	29857	27311	27074	27093	0.19	-
X-n242-k48	82751	85521	83300	83144	83040	0.35	+
X-n247-k47	37274	40870	39045	37681	37684	1.10	≈
X-n251-k28	38684	40576	39082	38916	38859	0.45	+
X-n256-k16	18839	20738	18888	18890	18889	0.27	+
X-n261-k13	26558	28631	26686	26672	26661	0.39	+
X-n266-k58	75478	78982	76022	75919	75757	0.37	+
X-n270-k35	35291	37130	35469	35451	35415	0.35	+
X-n275-k28	21245	22471	21245	21280	21254	0.04	+
X-n280-k17	33503	36313	33859	33689	33675	0.51	+
X-n284-k15	20215	22263	20380	20388	20390	0.86	≈
X-n289-k60	95151	98346	95939	95971	95691	0.57	+
X-n294-k50	47161	48487	47399	47360	47364	0.43	≈
X-n298-k31	34231	36317	34361	34371	34349	0.34	+
X-n303-k21	21736	23714	21920	21877	21871	0.62	≈
X-n308-k13	25859	28555	26121	26166	26081	0.86	+
X-n313-k71	94043	97700	94930	94919	94647	0.64	+
X-n317-k53	78355	79635	78400	78413	78398	0.05	+
X-n322-k28	29834	31862	30018	30038	29974	0.47	+
X-n327-k20	27532	29939	27682	27667	27629	0.35	+
X-n331-k15	31102	34351	31138	31178	31135	0.11	+

GPGLS-Experiments

The effectiveness of the GLS:

Both GPGLS and KGLS obtained **much better solutions than the CW heuristic**.

GPGLS managed to learn significantly **more effective GLS heuristics** than KGLS.

The effectiveness of utility function:

Both KGLS and GPGLS **outperformed the standard GLS** in lots of instances.

The **promising test performance** shows that GPGLS is able **to learn from small instances** and scale up to large instances.

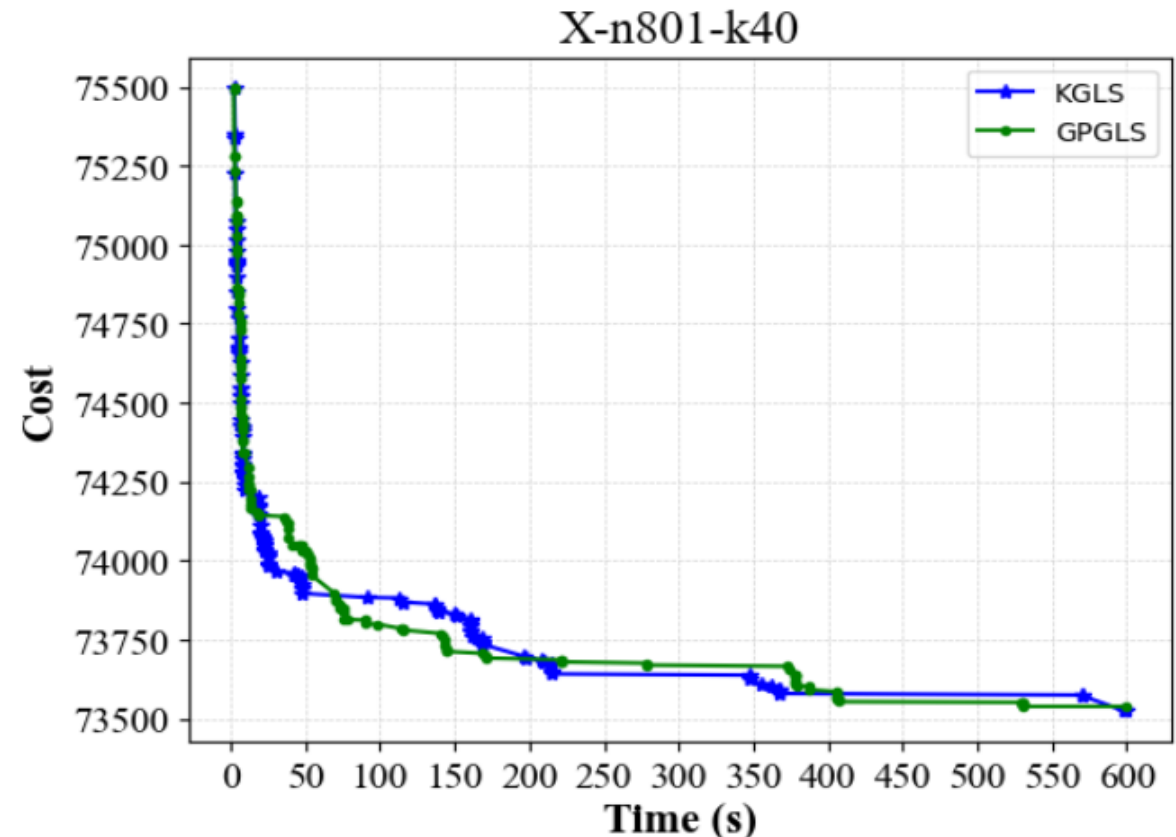
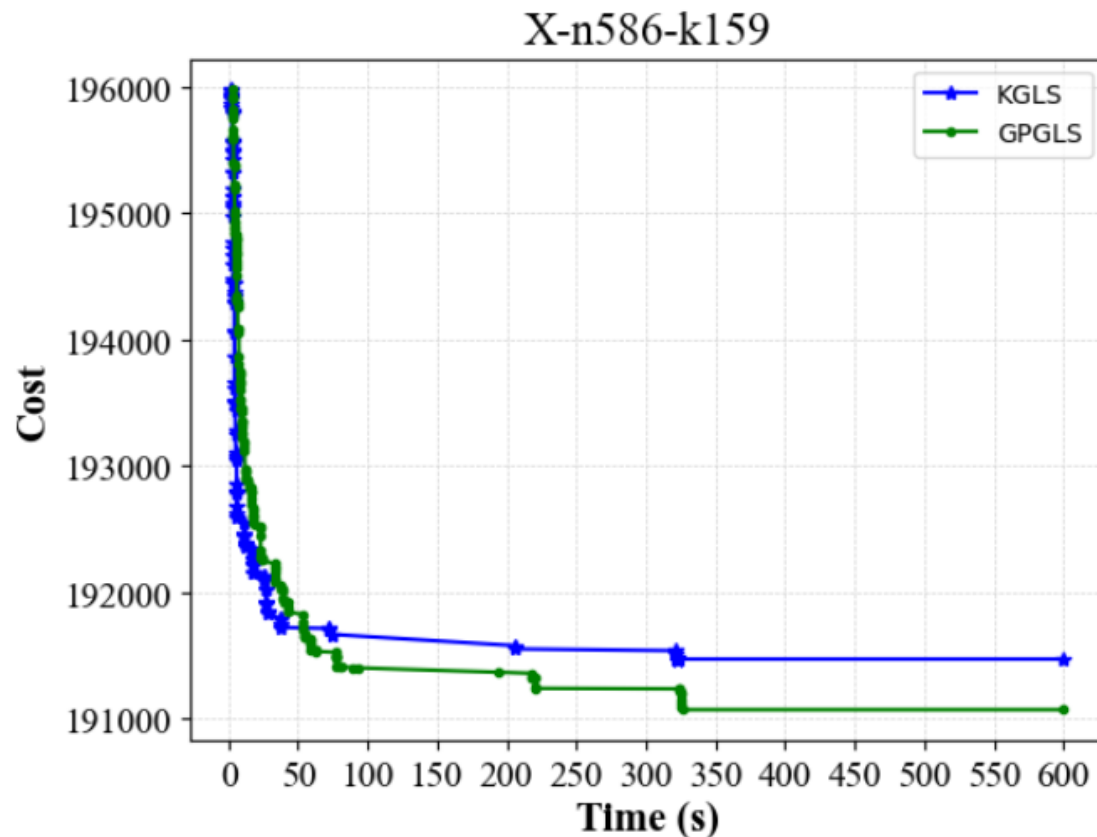
GPGLS-Analysis

Analysis

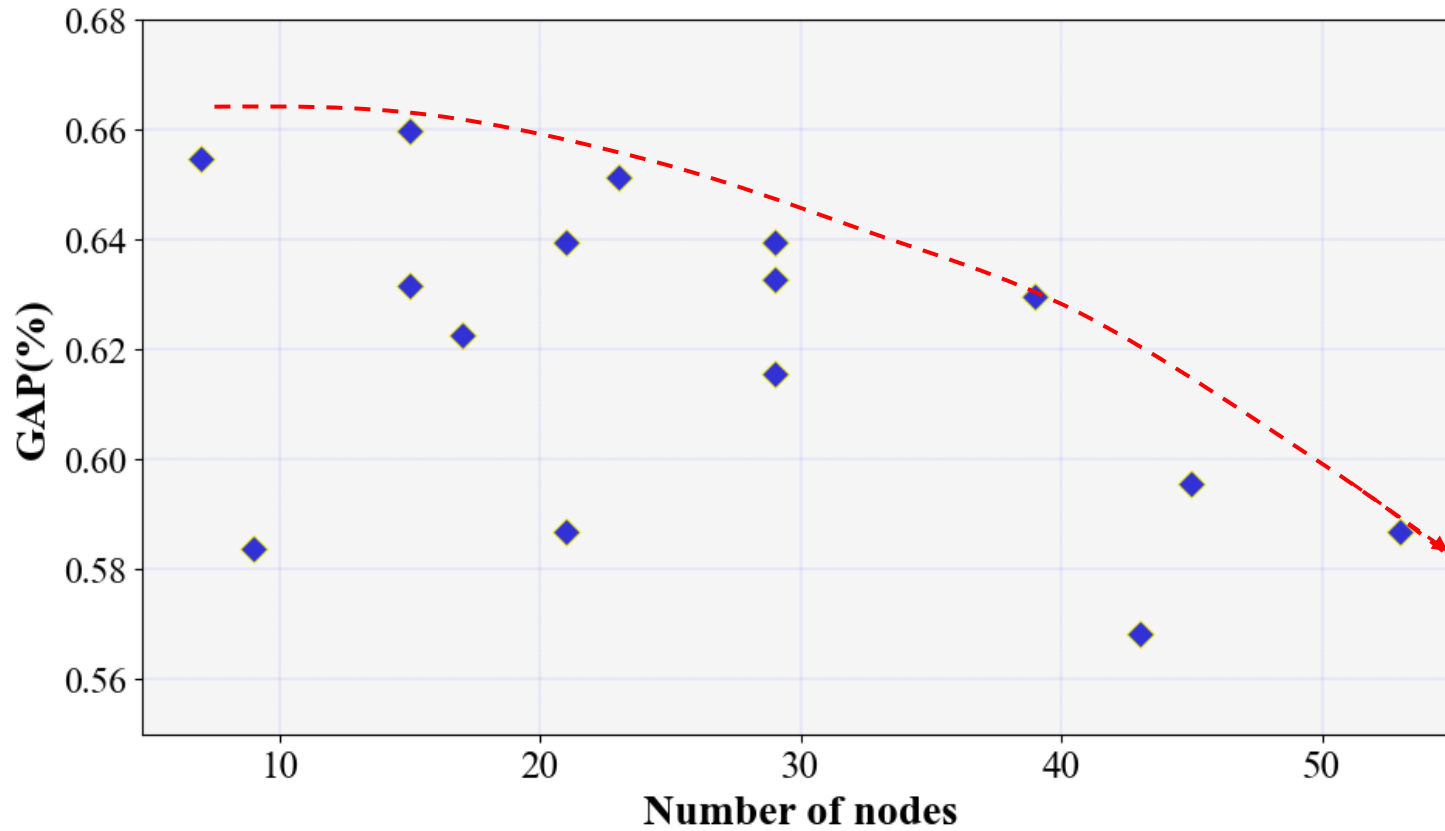
The **convergence curves** of KGLS and the GPGLS.

The behaviours **vary based on the characteristics** of the problem instance.

5 seconds for GLS during the training process **can be misleading in certain condition.**



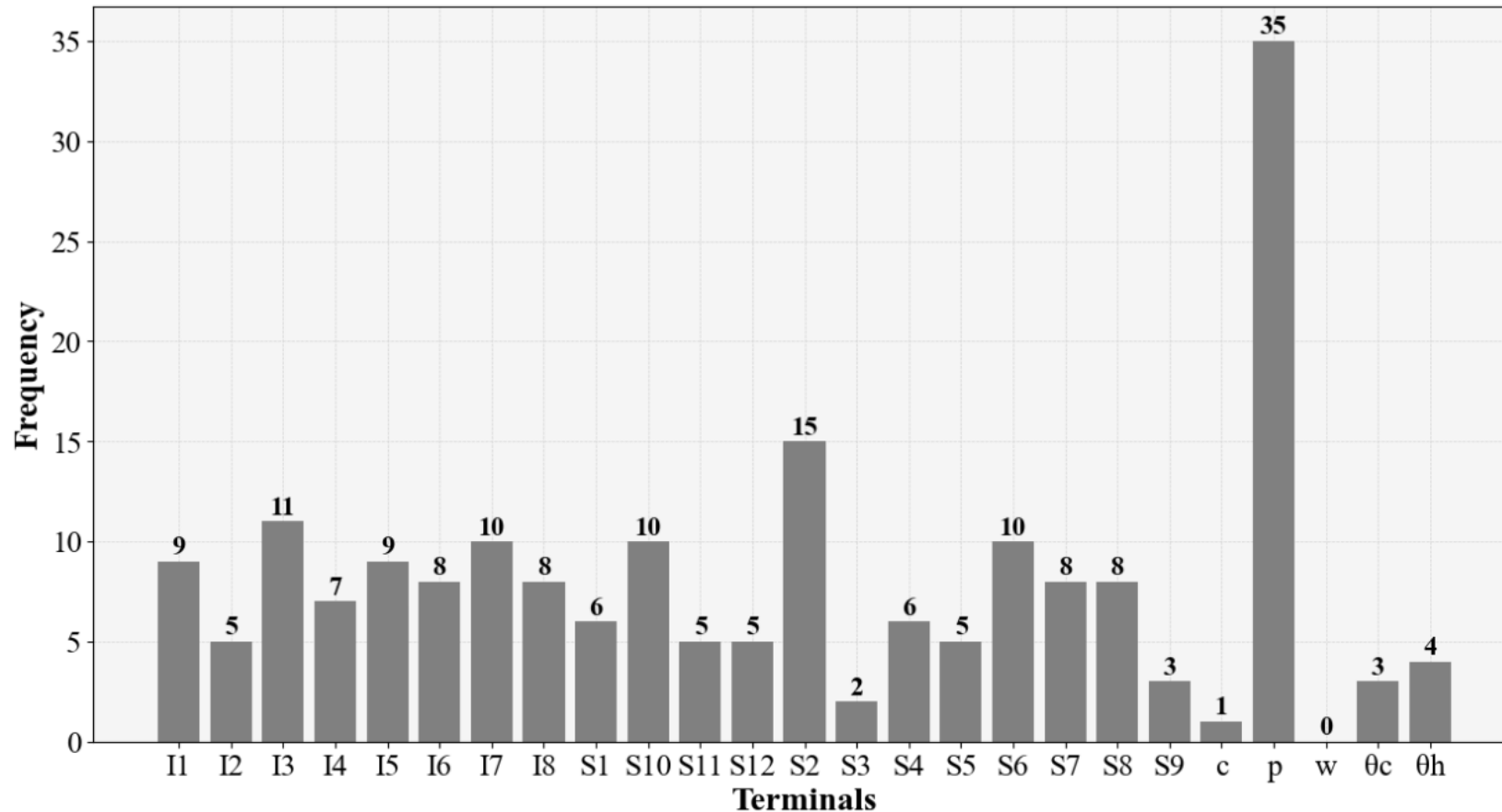
Length Analysis



The gap **tends to decrease** with the increase of the number of nodes, which is **consistent with intuition**.

This is **quite different** from the manual utility function of GLS and KGLS.

Features Analysis



KGLS: *c*, *p* and *S5* (width).

The rank of GPGLS:

p: numbers of penalized

S2: longest edge

I3: degree of capacity
utilization.

This implies the potential of
**completely new and
different patterns** from the
manually discovered ones.

Conclusion of Preliminary Work

- The goal: using GP to automatically design utility function for GLS.
- It has been successfully achieved by designing the new GPGLS framework with a comprehensive terminal set and a GLS-based fitness evaluation scheme.
- Further analysis demonstrates that GP managed to find completely new and different patterns from the utility functions used by existing GLS and KGLS methods.
- In general, GPGLS is a promising framework for automatically designing utility functions for GLS.

THANK YOU!

Q & A

Presenter: Saining Liu

Email: saining.liu@vuw.ac.nz



LLM Questions

1. The flowchart.
2. How to define which part should we use LLM.
3. How to tell LLM what you want to do? How to communicate with
4. The input and output.
5. What program language do you use?
6. How to evaluate and make comparison?
7. Could we use the LLM to represent individuals or population?
8. The slides and codes.